

6.24 (a) In the absence of electric moments, the Jefimenko expression for electric field becomes

$$\vec{E}(\vec{x}, t) = -\frac{1}{4\pi\epsilon_0} \int \frac{1}{c^2 R} \left[\frac{\partial \vec{J}(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} d^3x' = -\frac{1}{4\pi\epsilon_0} \int \frac{1}{c^2 R} \frac{\partial}{\partial t} [\vec{J}(\vec{x}', t)]_{\text{ret}} d^3x'$$

Expand $\frac{1}{R}$ to the first order in $\vec{x}$, we have $\frac{1}{R} = \frac{1}{r} + \frac{\vec{x} \cdot \vec{x}'}{r^3}$, with $r = |\vec{x}|$. We can also

approximate $t' = t - |\vec{x} - \vec{x}'|/c$ by $t' = t - r/c$. Then,

$$\begin{aligned} \vec{E}(\vec{x}, t) &= -\frac{1}{4\pi\epsilon_0} \int \frac{1}{c^2} \left(\frac{1}{r} + \frac{\vec{x} \cdot \vec{x}'}{r^3} \right) \frac{\partial}{\partial t} \vec{J}(\vec{x}', t - r/c) d^3x' \\ &= -\frac{\mu_0}{4\pi} \frac{\partial}{\partial t} \left(\int \frac{\vec{J}(\vec{x}', t - r/c)}{r} d^3x' + \frac{\vec{x}}{r^3} \cdot \int \vec{x}' \vec{J}(\vec{x}', t - r/c) d^3x' \right) \end{aligned}$$

The first term is zero, and the second term can be evaluated with the identity

$$\vec{x} \cdot \int \vec{x}' \vec{J}(\vec{x}') d^3x' = -\frac{1}{2} \vec{x} \times \int (\vec{x}' \times \vec{J}(\vec{x}')) d^3x'. \text{ Defined } \vec{m} = \frac{1}{2} \int \vec{x}' \times \vec{J}(\vec{x}') d^3x', \text{ we can write the}$$

electric field as

$$\vec{E}(\vec{x}, t) = -\frac{\mu_0}{4\pi} \frac{\partial}{\partial t} \left(-\frac{\vec{x}}{r^3} \times \vec{m}(t - r/c) \right) = \frac{\mu_0}{4\pi} \frac{1}{r^2} \hat{r} \times \frac{\partial \vec{m}(t - r/c)}{\partial t}, \text{ where } \hat{r} = \vec{x}/r.$$

Similarly, for the magnetic field,

$$\vec{B}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int \left\{ \left[\vec{J}(\vec{x}', t') \right]_{\text{ret}} \times \frac{\hat{R}}{R^2} + \left[\frac{\partial \vec{J}(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} \times \frac{\hat{R}}{cR} \right\} d^3x',$$

we have

$$\begin{aligned} [\vec{J}(\vec{x}', t')]_{\text{ret}} &= \vec{J}(\vec{x}', t - \frac{|\vec{x} - \vec{x}'|}{c}) = \vec{J}(\vec{x}', t - \frac{1}{c}(r - \hat{r} \cdot \vec{x}')) \\ &= \vec{J}(\vec{x}', t - r/c) + \frac{\partial \vec{J}}{\partial t}(\vec{x}', t - r/c) \frac{\hat{r} \cdot \vec{x}'}{c}, \end{aligned}$$

$$\frac{\hat{R}}{R^2} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} = (\vec{x} - \vec{x}') \left(\frac{1}{r^3} + \frac{3\vec{x} \cdot \vec{x}'}{r^5} \right) = \frac{\vec{x}}{r^3} + \frac{3\hat{r}(\hat{r} \cdot \vec{x}')}{r^3} - \frac{\vec{x}'}{r^3},$$

$$\frac{\hat{R}}{R} = \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^2} = (\vec{x} - \vec{x}') \left(\frac{1}{r^2} + \frac{2\vec{x} \cdot \vec{x}'}{r^4} \right) = \frac{\vec{x}}{r^2} + \frac{2\hat{r}(\hat{r} \cdot \vec{x}')}{r^2} - \frac{\vec{x}'}{r^2},$$

$$\text{and } [\vec{J}(\vec{x}', t')]_{\text{ret}} \times \frac{\hat{R}}{R^2} + \left[\frac{\partial \vec{J}(\vec{x}', t')}{\partial t'} \right]_{\text{ret}} \times \frac{\hat{R}}{cR}$$

$$= \left(\vec{J}(\vec{x}', t - r/c) + \frac{\partial \vec{J}}{\partial t}(\vec{x}', t - r/c) \frac{\hat{r} \cdot \vec{x}'}{c} \right) \times \left(\frac{\vec{x}}{r^3} + \frac{3\hat{r}(\hat{r} \cdot \vec{x}') - \vec{x}'}{r^3} \right) + \frac{\partial \vec{J}}{\partial t}(\vec{x}', t - r/c) \times \frac{1}{c} \left(\frac{\vec{x}}{r^2} + \frac{2\hat{r}(\hat{r} \cdot \vec{x}') - \vec{x}'}{r^2} \right)$$

$$\begin{aligned}
&= \ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \frac{\hat{\mathbf{r}}}{r^2} + \frac{1}{c} \frac{\partial \ddot{\mathbf{J}}}{\partial t}(\vec{x}', t-r/c) \times \frac{\hat{\mathbf{r}}}{r} \\
&\quad + \ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \vec{x}') - \vec{x}'}{r^3} + \frac{1}{c} \frac{\partial \ddot{\mathbf{J}}}{\partial t}(\vec{x}', t-r/c) \times \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \vec{x}') - \vec{x}'}{r^2} \\
&= \ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \frac{\hat{\mathbf{r}}}{r^2} + \frac{1}{c} \frac{\partial \ddot{\mathbf{J}}}{\partial t}(\vec{x}', t-r/c) \times \frac{\hat{\mathbf{r}}}{r} \\
&\quad + \left(1 + \frac{r}{c} \frac{\partial}{\partial t}\right) \ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \vec{x}') - \vec{x}'}{r^3}
\end{aligned}$$

Upon integration, the first two terms will drop out. For the third term, we have

$$\int \ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \vec{x}' d^3x' = -2\vec{m}$$

and $\int (\ddot{\mathbf{J}}(\vec{x}', t-r/c) \times \hat{\mathbf{r}})(\hat{\mathbf{r}} \cdot \vec{x}') d^3x' = \left(\hat{\mathbf{r}} \cdot \int \vec{x}' \ddot{\mathbf{J}}(\vec{x}', t-r/c) d^3x'\right) \times \hat{\mathbf{r}}$

$$= -\frac{1}{2} \left(\hat{\mathbf{r}} \times \int \vec{x}' \times \ddot{\mathbf{J}}(\vec{x}', t-r/c) d^3x' \right) \times \hat{\mathbf{r}} = -(\hat{\mathbf{r}} \times \vec{m}) \times \hat{\mathbf{r}} = \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \vec{m}) - \vec{m}.$$

Therefore,

$$\begin{aligned}
\vec{B} &= \frac{\mu_0}{4\pi} \frac{1}{r^3} \left(1 + \frac{r}{c} \frac{\partial}{\partial t}\right) \left[3(\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \vec{m}) - \vec{m}) - (-2\vec{m}) \right] \\
&= \frac{\mu_0}{4\pi} \frac{1}{r^3} \left(1 + \frac{r}{c} \frac{\partial}{\partial t}\right) \left[3(\vec{m}(t-r/c) \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \vec{m}(t-r/c) \right]
\end{aligned}$$

(b) For the configuration shown on the right, at z , we have

$$\vec{m}(t-r/c) = \pi a^2 N I (t - \sqrt{p^2 + z^2}/c) \hat{z},$$

and $\hat{\mathbf{r}} = \hat{z} \cos\theta + \hat{\rho} \sin\theta$, where $\sin\theta = \rho/\sqrt{p^2 + z^2}$. Using the

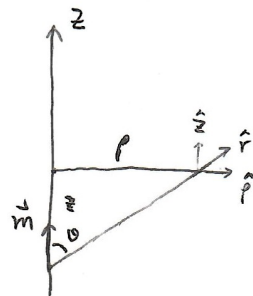
result from part (a), the electric field is given by

$$\vec{E} = \frac{\mu_0}{4\pi} \pi a^2 N \int_{-\infty}^{+\infty} \frac{1}{r^2} \frac{\partial I}{\partial t} (t - \sqrt{p^2 + z^2}/c) \hat{\mathbf{r}} \times \hat{z} dz.$$

$$\vec{E} = \frac{\mu_0}{4\pi} \pi a^2 N \frac{\partial I}{\partial t} (t - \rho/c) \left(\int_{-\infty}^{+\infty} \frac{\sin\theta}{r^2} dz \right) \hat{\phi}$$

$$= -\frac{\mu_0}{4\pi} \pi a^2 N \frac{\partial I}{\partial t} (t - \rho/c) \left(\int_{-\infty}^{+\infty} \frac{\rho}{(z^2 + \rho^2)^{3/2}} dz \right) \hat{\phi} = -\frac{\mu_0}{2} \frac{Na^2}{\rho} \frac{\partial I(t - \rho/c)}{\partial t} \hat{\phi},$$

where we have used the identity that $\int_{-\infty}^{+\infty} \frac{dz}{\sqrt{z^2 + \rho^2}} = \frac{z}{\rho^2}$.



We can perform similar calculation for the magnetic field. Notice that

$$3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m} = m[(3\cos^2\theta - 1)\hat{z} + 3\sin\theta\cos\theta\hat{\rho}]$$

then
$$\vec{B} \propto \frac{\mu_0}{4\pi} \pi a^2 N \int_{-\infty}^{+\infty} \frac{1}{r^3} \left(1 + \frac{r}{c} \frac{\partial}{\partial t}\right) I(t - r/c) [(3\cos^2\theta - 1)\hat{z} + 3\sin\theta\cos\theta\hat{\rho}] dz$$

There will be no $\hat{\rho}$ -component, as $\cos\theta = z/\sqrt{\rho^2 + z^2}$ is odd in z . Then,

$$\begin{aligned} \vec{B} &= \frac{\mu_0}{4\pi} \pi a^2 N \int_{-\infty}^{+\infty} \left[I(t - r/c) \frac{3\cos^2\theta - 1}{r^3} \hat{z} + \frac{\partial I(t - r/c)}{\partial t} \frac{1}{cr^2} (3\cos^2\theta - 1) \hat{z} \right] dz \\ &= \frac{\mu_0}{4\pi} \pi a^2 N \left[I(t - r/c) \int_{-\infty}^{+\infty} \frac{2z^2 - \rho^2}{(\rho^2 + z^2)^{5/2}} dz + \frac{\partial I(t - r/c)}{\partial t} \frac{1}{c} \int_{-\infty}^{+\infty} \frac{2z^2 - \rho^2}{(\rho^2 + z^2)^2} dz \right] \hat{z} \end{aligned}$$

The first integral is zero, as

$$\int_{-\infty}^{+\infty} \frac{2z^2 dz}{(\rho^2 + z^2)^{5/2}} = \int_{-\infty}^{+\infty} \frac{\rho^2}{(\rho^2 + z^2)^{5/2}} dz = \frac{4}{3\rho^2}$$

The second term can be written as

$$I(t - r/c) \cdot \frac{\partial I(t - r/c)/\partial t}{I(t - r/c)} \cdot \frac{1}{c} \int_{-\infty}^{+\infty} \frac{3\cos^2\theta - 1}{\rho r^2} dz$$

which can be ignored, as the condition suggests that

$$\frac{\partial I/\partial t}{I} \cdot \frac{1}{c} \ll 1$$

Therefore, $\vec{B} = 0$. In cylindrical coordinates,

$$\vec{\nabla} \times \vec{E} = -\hat{\rho} \frac{\partial E_\phi}{\partial z} + \hat{z} \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho E_\phi) = 0 \left(\frac{d^2 I}{dt^2} \right),$$

which satisfies the Faraday's law.